AGV Integrated Control System Interface Protocol

V3.0.6

Preface

Revision History

Version	Revision Content	Release Time	Applicable Version
V3.0.6	1. Changed the table of contents of the document 2. Added the schematic diagram of interaction flow and scene description 3. Added task list getting interface 4. Added map data getting interface 5. Added the order ID in the device information acquisition interface 6. Added enumerated values for task status reporting: Picked and placed 7. Added sub-task related fields in "actively getting task status" 8. Added order ID and shelf code in device status reporting and device status acquisition	May 2023	ICS3.2 & RCS4.2 and later
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Section 1 Preface

1.1 Purpose

This document is prepared to establish the implementation specifications of the interface protocol between a third-party system and the ICS system and to be used as reference for development on the customer side.

1.2 Audiences

Customer business managers, engineers, customer engineers, and API callers.

Section 2 Technical Agreement

2.1 Used Techniques

- Data in the systems on both sides is in JSON format.
- Interface interaction between systems is implemented over the HTTP Post protocol.
- APIs are synchronously called.
- The UTF-8 character set are uniformly used for encoding and decoding.
- The interaction data must contain the content specified in this document. Some non-mandatory fields may not be passed in as agreed.
- Developers on both sides develop interfaces based on the JSON sample and API description documents provided by the other side.
- The port number is 7000 by default and the IP address is the IP address of the RCS server. The actual interface request address and port number depend on the site condition.

2.2 Definitions of Part of Critical Values

2.2.1 Description of Part of Critical Values of Request Parameters

Parameter Name	Required or Not	Type	Description
orderId	(Required	string	Order ID. One order ID is generated for each movement task. The two systems use this order ID for data interaction during movement. Not duplicate
modelProcessCode	(Required	string	Business process code. One code for one production business process name. **You can get the code by querying "Business Process Code" on the "Business Process Template" on the third-party RCS**
fromSystem	(Required	string	Source system. Customized by a third party system. **The ICS system is notified of this information**

2.2.2 Basic Structure of Return Values

Parameter	Required or Not	Description
code	(Required	Return code: 1000: Success Others: Failed
desc	(Required	Description of return code
data	(Required	Return content

2.2.3 Definitions of Result Codes

Result Code	Description
1000	Request succeeded
1001	Request failed
7201	Incorrect request parameter
2103	Task process template number does not exist
2107	The path set is null when a waypoint needs to be passed
2108	Incorrect path set
2111	Please do not repeatedly assign tasks
2115	Task completed
Others	Other errors

2.3 Description of Authentication Request

2.3.1 Brief Introduction to AGV Interface Authentication

- AGV interfaces are authenticated by WSSE.
- This authentication mode is disabled in an Intranet environment by default. An HTTP request with a JSON body, as the simplest method, may be used for a third party.
- In an Intranet environment, authentication must be enabled for necessary security verification and data interception. In this case, the authentication logic must be implemented for interface requests from a third party. Username and password of the AGV system are used to make requests.

2.3.2 Code Encryption Logic in WSSE Authentication

- Get authorization from the AGV system manager to get the userName and password.
- Get the timestamp in seconds, stTime.
- Get the "timestamp + password" string.
- Convert the string subject to SHA-1 encryption into Base64 format and then encode it to form a digest token.
- Assemble a WSSE string in the format of: "useName + ":" + stTime + ":" + digest" (the example of the assembled WSSE encrypted string: admin:1623993204:rlqQDuh38bIj4SeyG1t5t+mh8lw=
- Place the WSSE string into the Authorization in the header of the HTTP request.
- Make normal interface requests.

Section 3 Commonly Used Interfaces

Table 3-1 Brief instruction to commonly used interfaces of the ICS system

SN	Interface Description	Details	Remarks
1	Task assigning interface	A third party system requests the ICS system to assign a task	
2	Task continuing interface	When you configure the AGV business template in such a way that the AGV suspends in a subtask process for acknowledgment, the AGV will suspend when proceeding to this subtask and wait for acknowledgment. Upon acknowledgment, a third party system calls this interface to enable the AGV to continue following tasks	
3	Task canceling interface	The interface is called when a third party system cancels a task	

Table 3-2 List of interfaces of a third party system

SN	Interface Description	Details	Remarks
1	Task status receiving interface	The interface is called when the ICS system actively reports the task status issued to a third party system	Must be implemented

3.1 Interfaces Provided by the ICS System

3.1.1 Task Assigning Interface*

3.1.1.1 Interface Details

- Interface name: http://IP:7000/ics/taskOrder/addTask
- Function description:
  - ◇ This interface is generally used by a third party system to assign tasks to the AGV.
  - ◇ A waypoint name can be customized by a third party system.
  - ◇ The integrated control system can convert a waypoint into a point that can be used by the AGV by mapping.
- Interface protocols: HTTP
- Provider: ICS (integrated control system
- Caller: Third party system
- Application: Assignment of all tasks

3.1.1.2 Parameter Details

Table 3-3 Details of request parameters

Parameter type	Parameter Name		Field Type	Max. Length	Required or Not	Description
Request parameters	modelProcessCode		string	100	Yes	Business process code. One code for one production business process name. **You can get the code by querying "Business Process Code" on the "Business Process Template" on the third-party RCS**
	priority		int	1	No	Priority: 4: High 6: Medium 8: Low Tasks with higher priority are scheduled first. The default value is 6.
	fromSystem		string	50	Yes	Source system: A third party system for assigning tasks, customized by the third-party system, for example, MES and WMS.
	orderId		string	100	Yes	Order ID: Specified when a third party system assigns tasks. Ensure that each order ID is unique and not duplicated
	taskOrderDetail	taskPath	string	200	No	Collection of task waypoints: The starting point and destination point of AMR. Multiple points are separated by commas (,. When a task is assigned, you can pass in the region code maintained in the RCS, and then the RCS will select available waypoints from the incoming region. (the RCS is responsible for maintaining the bin information in each waypoint in the region.
		shelfNumber	string	32	No	Shelf code: Assign this code when a third party assigns a shelf if necessary
		assignRobotIds	string	64	No	Device SN or device ID. Devices

						are separated by commas. For detailed parameters, refer to Device Management -> AMR Management.
		releaseRobot	int	1	No	Determine whether to release AMR after the task is executed: Do not release AMR and release AMR. If the AMR is not released, you can specify the AMR to continue following tasks
		deviceNum	string	32	No	The numbering of the device executing this task. The device number is a 4-digit value that can be viewed from Device Management -> AMR Management. Only one AMR number can be passed in. Used only when the AMR is not released or for associated order
		loadType	int	2	No	You can customize AMR speed in the RCS system under different load states of the AGV Default load state: 1: Unloaded shelf 2: Half-loaded shelf 3: Full-loaded shelf
		shelfModel	int	2	No	Load type: 1: Square shelf 2: Rectangular shelf 3: Traction load 4: Pallet 7: Forklift workbin 8: Rectangular pallet
Response	code		int	6	Yes	Return status code
	data		string		Yes	Return result
	desc		string	64	Yes	Status code description

3.1.1.3 Example

- Full-parameter request (pass in required parameters and ignore unnecessary ones as needed

```
{
    "modelProcessCode": "chuku",
    "priority": 6,
    "fromSystem":"MES",
```

```
        "orderId":"3215435",
        "taskOrderDetail": [
            {
                "taskPath":"shelf area, worktable, shelf area",
                 "shelfNumber":"80000032"
            }
        ]
    }
```

- Successful response

```
{
   "code": 1000,
   "data": "3215435",
   "desc": "Request successfully"
}
```

- Response when the task process number is not configured

```
{
   "code": 2103,
   "data": "",
   "desc": "Task process template number does not exist"
}
```

- Description: When this error code is returned, the business process template number is not configured. The configuration item is displayed on the leftmost tab page in task template configuration.

- Response when an order ID is repeatedly issued

```
{
   "code": 2111,
   "data": "",
   "desc": "Please do not repeatedly assign tasks"
}
```

- Description: When this error code is returned, a task has already been assigned to the ICS and the ICS has written it to the database, but the task with this order ID is repeatedly assigned.

- Response when an error occurs when assigning task path:

```
{
   "code": 2108,
   "data": "",
   "desc": "Incorrect collection of task points"
}
```

- Description: When this error code is returned, The number of assigned points required by the RCS task template is inconsistent with the actual number of assigned points.

3.1.2 Task Continuing Interface*

3.1.2.1 Interface Details

- Interface name: http://IP:7000/ics/out/task/continueTask
- Function description: This interface is called when a task is suspended at a certain point. A third party system shall confirm and continue the task. The interface is triggered by passing in order ID or shelf code/AGV location/device ID. The priority of continuing the task is order ID -> Device ID -> AGV Location -> Shelf code. When both the order ID and device ID are passed in, the task with the order ID will first be triggered. When there are multiple areas in the RCS system, and there is a certain identical point number in these areas, it is necessary to use areaId to distinguish the task to be continued when the current AGV location (point number is passed in.
- Interface protocols: HTTP
- Provider: ICS (integrated control system
- Caller: Third party system
- Application: A task suspends at a certain point. The AGV continues the task by manual intervention. This can also guarantee safety before the AGV approaches the bin. When receiving status 10, the MES system will wait for safety acknowledgment. After checking that the AGV can safely approach the bin, the system can call this interface to let the AGV approach the bin as the basis for safety acknowledgment.

3.1.2.2 Parameter Details

Table 3-4 Parameter details of task continuing interface

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	orderId	string	100	No	(Trigger a task according to order ID from a third party system Order ID specified when a third party system assigns a task
	shelfNumber	String	32	No	(Trigger a task according to shelf code Shelf code
	deviceNum	String	32	No	(Trigger a task according to device ID Device ID

	currentStation	String	32	No	(Trigger a task according to current point Current point coordinates in QR code form. areaId must be passed in when this point ID exists in multiple areas.
	areaId	int	2	No	Area ID, passed in when there are multiple areas in the RCS system, and there is a certain identical point number in these areas
Response	code	int	6	Yes	Return status code
	desc	string	64	Yes	Status code description

3.1.2.3 Example

- Request

```
{
"orderId":"WenDingXingTaskTest514"
}
```

- Response

```
{
"code":1000,
"desc":"success"
}
```

3.1.3 Task Canceling Interface*

3.1.3.1 Interface Details

- Interface name: http://IP:7000/ics/out/task/cancelTask
- Function description: This interface is called when a third party needs to actively cancel a task. If the AGV is unloaded when the task is canceled, the AGV will be released. If the AGV is loaded, there are three handling methods when the task is canceled:
  - ◇ The AGV stands still and waits for manual intervention when the task is canceled. This is the default method.
  - ◇ The AGV moves the shelf or pallet to the starting point when the task is canceled. Configuration is required.
  - ◇ Specify a destination point when the task is canceled. The AGV will move the goods to the specified point. Configuration is required.

For details about configuration, in the RCS platform, locate Configuration Management -> Task Template Configuration -> RCS Task Template, and click Template No. to configure the task canceling policy under load condition.

- Interface protocols: HTTP
- Provider: ICS (integrated control system
- Caller: Third party system
- Application: The WMS and MES systems need to cancel tasks assigned to the RCS

3.1.3.2 Parameter Details

Table 3-5 Parameter details of task canceling interface

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	orderId	string	100	No	Order ID from a third party system. Tasks can be canceled in bulk. Data is passed in array format, as shown in the following example:
	deviceNumber	string	50	No	Device ID, similar to order ID from a third party system. Enter either device ID or order ID. Only order ID from a third party system can be canceled.
	destPosition	string	64	No	The destination point where the shelf is placed after the task is canceled. The shelf will be placed in the specified position only when it is lifted. Otherwise, the task will be directly canceled. If this value is not passed in, the shelf will be placed in its current position or moved to the starting position after the task is canceled.
Response	code	int	6	Yes	Return status code
	desc	string	64	Yes	Status code description

3.1.3.3 Example

- Request
- Cancel according to device ID

```
[
    {
        "deviceNumber": "wewf",
```

```
        "destPosition": "01001047"
    }
]
```


- Cancel according to order ID


```
[{
"orderId":"WenDingXingTaskTest514",
"destPosition":"point 1"
},
{
"orderId":"WenDingXingTaskTest512",
"destPosition":"point 2"
}]
```


- Description: Tasks can be canceled in bulk. destPosition is passed in only when the AGV is instructed to move the shelf to the destination point after the task is canceled.
- Response


```
{
"code":1000,
"desc":"success"
}
```


3.2 Interfaces Provided by Third Party System

3.2.1 AGV Notifies a Third Party System of Task Status*

3.2.1.1 Interface Details

- Interface name: Provided by a third party system. After an interface is determine, please immediately deliver the interface to the site implementation engineer. The interface shall be configured before task status can be reported to the third party system.
- Function description: This interface is to be provided by a third party system. The interface is called when the integrated control system actively report the task status. Some fields can be ignored and reserved fields are used for extended functions.
- Interface protocols: HTTP
- Provider: Third party system
- Caller: ICS (integrated control system
- Application: A third party system needs to receive the task status reported from the AGV. In a logic processing, you need to configure the RCS task template so that the task status can be reported to the third party system.

- Calling logic: A third party system provides an interface URL that is implemented based on the following protocol. At the node where task status needs to be reported to a third party system, the task template is configured in the RCS so that the task status can be reported to the third party system. When the AGV runs to the corresponding node, the task execution status will be reported to the third party system based on the protocol.
- Retry mechanism: If 1000 is not returned from the third party system, after receiving the request, the RCS will retry according to the retry setting until the setting is exceeded or the third party system successfully returns 1000.

3.2.1.2 Parameter Details

Table 3-6 Parameter details of task status interface

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	orderId	string	100	Yes	Order ID from a third party system
	qrContent	string	100	No	The destination point to which the current subtask will proceed, corresponding to the point assigned when a task is assigned
	deviceNum	string	32	No	AGV ID
	deviceCode	string	64	No	AGV SN
	status	int	2	Yes	Task status: 3: Canceled 5: Sending failed 6: Running 7: Execution failed 8: Completed 9: Assigned 10: Wait for acknowledgment 20: Picking 21: Picked 22: Placing 23: Placed
	storageNum	string	32	No	Bin ID, corresponding to bin location
	modelProcessCode	string	32	Yes	Task process template number
	errorDesc	string	64	No	Task execution failed, or cause of failure when sending to the AGV
	shelfNumber	string	32	No	Shelf code
	shelfCurrPosition	string	32	No	Current shelf position
	subTaskStatus	String	2	Yes	Action status of AGV in the RCS task template, extended field: 1: Not started 2: Running* 3: Completing* 4: Failed 5: Cancel

	subTaskTypeId	String	6	Yes	AGV action type, extended field
	subTaskId	String	10	Yes	RCS substask ID, reserved field
	subTaskSeq	String	2	Yes	Action number, starting from 0, extended field
	icsTaskOrderDetailId	String	10	Yes	Task ID registered in the ICS, extended field
Response	code	int	6	Yes	Return status code
	desc	string	64	Yes	Status code description

3.2.1.3 Example

- Request

```
{
    "storageNum": "C13",
    "subTaskStatus": "3",
    "orderId": "1637846849",
    "deviceCode": "7J08197 PA400001 ",
    "modelProcessCode ": "F3SensorTriggerTaskProcess ",
    "subTaskTypeId ": "20 ",
    "subTaskId ": "2553259 ",
    "deviceNum ": "dn19 ",
    "qrContent ": "33000513 ",
    "subTaskSeq ": "0 ",
    "icsTaskOrderDetailId ": "703303 ",
    "status ": 5,
"errorDesc ": "33000513 task point does not exist",
"shelfCurrPosition": "33000511"
}
```

- Successful response

```
{
"code":1000,
"desc":"success"
}
```

Section 4 Optional Interfaces

Table 4-1 Brief introduction to optional interfaces of the ICS system

SN	Interface Description	Details	Remarks
1	Query the task status	A third party system requests the ICS system to get the task status when it needs to actively get the task status	The ICS will actively reports the task status in normal cases
2	Change the destination point of a task	This interface is called when a third party system needs to change the destination point of a task after the task is assigned to the RCS.	
3	Rotate the shelf	A worker requests the AGV loaded with shelf to rotate the shelf when it stops at the worktable.	
4	Update task assignment score	After a task is assigned, AGVs wait in queue to move goods. This interface is called when manual intervention is required to adjust the task execution sequence.	
5	Get all bin information	Return empty and full bin status of all bins recorded by the RCS according to task information.	
6	Update bin status	Manually intervene to request the RCS to change the empty or full status of a bin.	
7	Get all shelf locations	Request the RCS to get all shelf locations in the current area when maintaining all shelf locations in the RCS.	
8	Update shelf location	This interface is called when the shelf is manually moved when the shelf is maintained in the AGV system.	
9	Get AGV status	This interface is called when a third party system needs to get AGV running status.	
10	Get the algorithm-based AGV programming status	This interface can be called to get AGV status when the AGV stops at a certain point.	
11	Terminate a task	This interface is called when a task must be directly terminated on the upper level to release the AMR when the task runs to a certain step.	
12	Cancel device preemption	Device exception occurs. After the device is manually restored from the exception, task information and shelf information bound to the device will be unbound. This process will be automatically canceled when the task is in progress.	
13	Suspend/recover device	This interface is used to suspend/recover a single or multiple devices.	
14	Get task path	This interface is called to get the path information and	

		locked block information in the current planned path of a certain AMR.	
15	Apply for entering the traffic control area	This interface is called when other manufacturer's AMR needs to pass the area where the current AMR is operating, the other manufacturer's AMR will make an application, and the RCS will start traffic control.	
16	Enable and disable the prohibited region	If the RCS system is configured with a prohibited region, the customer system can enable or disable the prohibited region.	
17	Pre-scheduling task	The AGV arrives at a point in advance when the customer system needs to issue a task in advance.	
18	Get the map	This interface is called to get the map file.	
19	Get the task template	This interface is called to get the name of the RCS task template and list of numbering configurations.	
20	Get the process template	This interface is called to get the name and numbering of the business process template and the list of numbering configurations.	

Table 4-2 List of interfaces of a third party system

SN	Interface Description	Details	Remarks
1	Report the device status	This interface is called when the ICS system actively reports device status to a third party system.	Implemented as needed
2	Get task points	The ICS calls this interface when the running AGV needs to get the next task point if no task point is specified when a third party system assigns a task.	Implemented as needed
3	Report alarm message	This interface is called to report AGV alarm messages.	Implemented as needed
4	Actively report shelf location to a third party system	This interface is called to actively report all shelf locations to a third party system.	Implemented as needed

4.1 Interfaces Provided by the ICS System

4.1.1 Task Related Interfaces

4.1.1.1 Query the task status

4.1.1.1.1 Interface Details

- Interface name: http://IP:7000/ics/out/task/getTaskOrderStatus

- Function description: This interface is called by a third party system to actively get AGV task status. This interface can be called to get current task status for all assigned tasks.
- Interface protocols: HTTP
- Provider: ICS (integrated control system
- Caller: Third party system
- Application: The task status can be actively obtained if the RCS has not reported the task status yet even after the tolerance time a system waits for the task to be completed.

4.1.1.1.2 Parameter Details

Table 4-3 Parameter Details

Parameter type	Parameter Name		Field Type	Max. Length	Required or Not	Description
Request	orderId		string	100	Yes	Order ID specified when a third party system assigns a task
Response	code		int	6	Yes	Return status code
	data		JsonObject	1000	Yes	Task status information returned
	data	areaId	int	10	Yes	Area ID: Provided by the integrated control system. This field is mainly used to distinguish areas or warehouses.
		createTime	long	12	Yes	Task creation time: Timestamp
		fromSystem	string	50	Yes	Source system: A third party system assigning tasks. Customized by a third party system.
		status	int	2	Yes	Task group status: 1: Not sent 3: Canceled 4: Sending 5: Sending failed 6: Running 7: Execution failed 8: Completed 9: Assigned 10: Wait for acknowledgment Deemed complete if all subtasks are completed
		taskOrderDetail	JsonArray	1000	Yes	Task status details
	taskOrderDetail	qrContent	string	100	No	Current task location

		deviceNum	string	32	No	Numbering of AGV in task
		status	int	2	Yes	Subtask status: 1: Not sent 3: Canceled 4: Sending 5: Sending failed 6: Running 7: Execution failed 8: Completed 9: Assigned 10: Wait for acknowledgment
		time	long	13	Yes	System time of current status
		subTaskStatus	String	2	Yes	Action status of AGV in the RCS task template, extended field 1: Not started 2: Running* 3: Completing* 4: Failed 5: Cancel
		subTaskTypeId	String	6	Yes	AGV action type, extended field, see Appendix 4 - Subtask Type Reference Table
		subTaskId	String	10	Yes	RCS substask ID, reserved field
		subTaskSeq	String	2	Yes	Action number, starting from 0, extended field
		subTaskStatus	String	2	Yes	Action status of AGV in the RCS task template, extended field 1: Not started 2: Running* 3: Completing* 4: Failed 5: Cancel
	desc		string	64	Yes	Status code description

4.1.1.1.3 Example

- Request

```
{
    "orderId":"WMS1100223"
```

```
}
```

- Response

```
{
  "code": 1000,
  "data": {
      "areaId": 1,
      "createTime": 1684920666,
      "taskOrderDetail": [
          {
              "deviceNum": "ffff",
              "qrContent": "0040",
              "subTaskStatus": 3,
              "subTaskSeq": 3,
              "time": 1684920709,
              "deviceCode": "ffff0000",
              "subTaskTypeId": 5,
              "subTaskId": 4,
              "status": "8"
          }
      ],
      "fromSystem": "call_task",
      "status": 8
  },
  "desc": "Request successfully"
}
```

4.1.1.2 Change Task Point

4.1.1.2.1 Interface Details

- Interface name: http://IP:7000/ics/out/task/updateOrderPointInfo
- Function description: This interface is called if the starting and destination points need to be changed when the AGV has not arrived at the destination point yet after a task is assigned, in a movement task, for example, changing point A→B to C→D.
- Interface protocols: HTTP
- Provider: ICS (integrated control system
- Caller: Third party system

- Application: A task has been assigned but the destination point needs to be changed due to changes in business requirements.

4.1.1.2.2 Parameter Details

Table 4-4 Parameter details of task continuing interface

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	orderId	string	100	Yes	Order ID specified when a third party system assigns a task
	pointPath	string	255	Yes	Task waypoint. Both the starting and destination points after change shall be inherited.
Response	code	int	6	Yes	Return status code
	desc	string	64	Yes	Status code description

4.1.1.2.3 Example

- Request

```
{
"orderId":"WenDingXingTaskTest514",
"pointPath":"80000231,80000212"
}
```

- Description:

  1. Change to a task has certain limitations. For example, for point A -> point B, when the AGV arrives at point A, the task can no longer be changed to C -> D, but can be changed to A -> D, and when the AGV arrives at point B, the task point cannot be changed any longer for this task.
  2. In simple terms, after a task is assigned, when the AGV arrives at a point, the task point can no longer be changed.
  3. When a task point is changed, the starting and destination points shall be inherited throughout the entire path.

- Successful response

```
{
"code":1000,
"desc":"success"
}

```

- Response when order information does not exist

```
{
"code":3003,
"data":"order detail is lost",
```

- "desc":"failed"
- }

4.1.1.3 Update Task Priority

4.1.1.3.1 Interface Details

- Interface name: http://IP:7000/ics/out/updateTaskPriority
- Function description: This interface is called when manual intervention is required to adjust the task execution sequence, and this task prioritizes in AGV assignment if no AGV is assigned to execute the assigned task. Assign an AGV to execute this task.
- Interface protocols: HTTP
- Provider: ICS (integrated control system
- Caller: Third party system
- Application: There are too many tasks and AGVs, and some task sequences should be manually adjusted.

4.1.1.3.2 Parameter Details

Table 4-5 Parameter details of task priority updating interface

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	orderId	string	100	Yes	Order ID specified when a third party system assigns a task
	score	string	10	Yes	Range: 500 to 100000. The request will be executed immediately in a single increase of 100,000. The score is a AGV assignment priority score calculated by the algorithm according to the distance from the starting point and the time weight. The higher the score, the higher the priority.
Response	code	int	6	Yes	Return status code
	desc	string	64	Yes	Status code description

4.1.1.3.3 Example

- Request

- {
- "orderId":"chengpinchukufenjian01_20",
- "score":"100000"

- }
  - Note: This interface is useful before an AGV is assigned to execute this task
    - Response

- {
- "code":1000,
- "desc":"success"
- }

4.1.1.4 Terminate a Task

4.1.1.4.1 Interface Details

- Interface name: http://IP:7000/ics/out/endTask
- Function description: This interface called by a third-party system to terminate a task in advance.
- Interface protocols: HTTP
- Provider: ICS (integrated control system
- Caller: Third party system
- Application: Applicable to scenarios where a third party system wants to terminate a task in advance. The third party system assigns a temporary task to ask the AGV to proceed to the next point until the system actively terminates the task in advance when no longer wanting the AGV to continue the task.

4.1.1.4.2 Parameter Details

Table 4-6 Parameter details of task termination interface

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
	orderId	string	100	Yes	Order ID: To be specified when a third party system terminates a task. Pass in this order ID when a third party system terminates a task.
Response	code	int	6	Yes	Return status code
	data	string		Yes	Return result
	desc	string	64	Yes	Status code description

4.1.1.4.3 Example

- Request

- {
- "orderId":"3215435"
- }
  - Response

- {

```
    "code": 1000,
    "data": "",
    "desc": "Request successfully"
}
```

4.1.1.5 Pre-scheduling task

4.1.1.5.1 Interface Details

- Interface name: http://IP:7000/ics/taskOrder/preAllocTask
- Function description: This interface is called by the customer system to assign tasks in advance.
  - ◇ For example, there will be a pallet of goods to be loaded in worktable A after one minute, this interface can be called to assign a task to the AGV to arrive at point A, and the AGV will arrive at point A within one minute.
- Interface protocols: HTTP
- Provider: ICS (integrated control system
- Caller: Third party system

4.1.1.5.2 Parameter Details

Table 4-7 Parameter details of pre-scheduling task

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	webCode	string	10	Yes	Device ID
	taskCountDown	int	4	No [Do not left blank if possible]	Task countdown, in seconds
	taskTyp	string	10	No [Do not left blank if possible]	Task process template number
Response	code	int	6	Yes	Returned status code, error when no 1000 is returned
	desc	string	64	Yes	Status code description
	data	JSON	64	No	

4.1.1.5.3 Example

- Request

```
[
{
"webCode": "07170119",
        "taskCountDown":100,
"taskTyp":"move"
```

```
}
]
```

- Response

```
{
  "code": 1000,
"desc": "Request successfully"
}
```

4.1.1.6 Shelf Rotation Interface

4.1.1.6.1 Interface Details

- Interface name: http://IP:7000/ics/out/changeShelfSide
- Function description: In scenarios where picking is required, when the AGV loaded with shelf stops at the worktable and waits for the continue instruction while a worker is picking goods and may be inconvenient to pick goods on the other side, the interface can be called by the RCS to make the AGV rotate 180 degrees.
- Interface protocols: HTTP
- Provider: ICS (integrated control system
- Caller: Third party system
- Application: Applicable to goods picking in warehouses to facilitate manual picking.

4.1.1.6.2 Parameter Details

Table 4-8 Parameter details of shelf rotation interface

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	orderId	string	100	No	Order ID specified when a third party system assigns a task
	shelfNumber	String	12	No	Shelf code. Use either orderId or shelf code. orderId shall prevail when both parameters are passed in.
	shelfToSide	string	64	Yes	Shelf orientation after rotation A: Side A orients to work position B: Side B orients to work position The angle of side A or side B of the shelf relative to this worktable shall be configured on the RCS

	toLocationCode	string	64	Yes	Picking worktable to go to after rotation
Response	code	int	6	Yes	Return status code
	desc	string	64	Yes	Status code description

4.1.1.6.3 Example

- Request

```
{
    "orderId":"chengpinchukufenjian01_20",
    "shelfToSide":"A",
    "toLocationCode":"81001945"
}
```

- Description: This function is can be used only when picking is required and the angle of side A/B relative to the point should be configured on the RCS.
- Response

```
{
"code":1000,
"desc":"success"
}
```

4.1.1.7 Query Task List

4.1.1.7.1 Interface Details

- Interface name: http://IP:7000 /ics/out/task/getOrderList
- Function description: Query the current task information in the specified area.
- Interface protocols: HTTP
- Provider: ICS (integrated control system
- Caller: Third party system
- Application: Used when it is necessary to get the specific or all task information.

4.1.1.7.2 Parameter Details

Table 4-9 Parameter details of task list querying interface

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	areaId	int	100	Yes	Area ID
	templateCode	String	12	No	Task template number. Multiple numbers are separated by commas.
	OrderStatus	String	64	No	Task status:

							1: Not sent 2: Canceling 3: Canceled 4: Sending 5: Sending failed 6: Executing 7: Execution failed 8: Task completed 9: Assigned Items are separated by commas.
	Priority				String	64	No	Priority: 4: Low 6: Medium 8: High
	SearchKey				String	64	No	Device ID, single choice
	qrName				String	64	No	Node name/point QR code, single choice
	shelfNum				String	64	No	Shelf code, single choice
	storageNum				String	64	No	Bin ID, single choice
	orderIdList				JsonArray	64	No	Order ID list. Order IDs are of string type and separated by commas.
	taskIdList				JsonArray	64	No	Task ID list. Task IDs are of string type and separated by commas.
	batchIdList				JsonArray	64	No	Batch ID, for customization purpose
	intendStartTimeHigh				String	64	No	Max. filter time
	intendStartTimeLow				String	64	No	Min. filter time
	StartPage				String	64	No	Number of starting pages
	pageSize				String	64	No	Entries per page
Response	code				int	6	Yes	Return status code
	desc				String	64	Yes	Status code description
	data				JsonObject	64	Yes	Returned information
	data	Tasks			JsonArray	64	Yes	Task list array
			Tasks	AllowChangeDevice	int	2	Yes	Allow changing AGV or not: 0: Not allowed 1: Allowed
				RobotType	String	64	Yes	Device type

			TaskId	int	2	Yes	Task ID
			ZoneId	Integer	2	Yes	Group ID
			Priority	int	2	Yes	Priority: 4: Low 6: Medium 8: High
			OrderPath	String	64	Yes	Task path
			OrderNumber	String	64	Yes	outOrderId of taskGroup table
			OrderId	String	64	Yes	thirdOrderId of taskGroup table
			Material	String	64	No	Material information
			TemplateCode	String	64	Yes	Task template number
			ReExecute	int	64	Yes	Allow re-executing or not: 0: Not allowed 1: Allowed
			OrderStatus	int	64	Yes	Order status
			createTime	String	64	Yes	Creation time
			TemplateName	String	64	Yes	Task template name
			TaskTime	String	64	Yes	Extended parameter, unused
		TotalPage		int	4	Yes	Total pages
		pageSize		int	4	Yes	Max. number of entries per page
		Page		int	4	Yes	Current number of pages
		totalCount		int	4	Yes	Total counts

4.1.1.7.3 Example

- Request

- Description: Query the list of all tasks without filter conditions

```
{
"areaId": 1
}
```

- Filter out tasks meeting filtering conditions

```
{
    "areaId": 1,
    "templateCode": "templateCode1",
    "OrderStatus": "6",
    "Priority": "6",
    "SearchKey": "robot-1",
    "qrName": "00000183",
    "shelfNum": "shelf0",
    "storageNum": "H001",
    "orderIdList": ["103","104"],
    "taskIdList": ["2101","2102"],
    "batchIdList": [],
    "intendStartTimeHigh": "2023-05-24 23:59:59",
    "intendStartTimeLow": "2023-05-23 00:00:00",
    "StartPage": "1",
    "pageSize": "20"
}
```


- Response


```
{
    "code": 1000,
    "data": {
        "Tasks": [
            {
                "AllowChangeDevice": 0,
                "RobotType": "DH-RTA-C060-L-2L",
                "TaskId": 2102,
                "ZoneId": 1,
                "Priority": 6,
                "OrderPath": "33000204—00000183",
                "OrderNumber": "1968",
                "OrderId": "e780893b-efb1-405c-8078-b4efd99cf548_1_6990",
                "Material": "[]",
                "TemplateCode": "banhuojiaxiaxian1",
                "ReExecute": 1,
```

```
                "OrderStatus": 9,
                "createTime": "2023-05-24 10:16:22",
                "TemplateName": "Shelf movement area handover",
                "TaskTime": "— —"
            },
            {
                "AllowChangeDevice": 0,
                "RobotType": "DH-RTA-C060-L-2L",
                "TaskId": 2101,
                "ZoneId": 1,
                "Priority": 6,
                "OrderPath": "33000204—00000183",
                "OrderNumber": "1967",
                "OrderId": "415a94d2-f419-4532-a4d9-940f3bd062cd_1_6075",
                "Material": "[]",
                "TemplateCode": "banhuojiaxiaxian1",
                "ReExecute": 1,
                "OrderStatus": 9,
                "createTime": "2023-05-24 10:13:52",
                "TemplateName": "Shelf movement area handover",
                "TaskTime": "— —"
            }
        ],
        "TotalPage": 1,
        "pageSize": 20,
        "Page": 1,
        "totalCount": 2
    },
    "desc": "Request successfully"
}
```

4.1.2 Device Related Interface

4.1.2.1 Get AGV status

4.1.2.1.1 Interface Details

- Interface name: http://IP:7000/ics/out/device/list/deviceInfo
- Function description: Get AGV running status information, such as AGV location, device ID, device SN, and device status. This interface can be used to set a data dashboard
- Interface protocols: HTTP
- Provider: ICS (integrated control system
- Caller: Third party system
- Application: When a third party system needs to set a data dashboard

4.1.2.1.2 Parameter Details

Table 4-10 Parameter details of AGV status getting interface

Parameter type	Parameter Name		Field Type	Max. Length	Required or Not	Description
Request	areaId		string	10	Yes	Area ID: Provided by the integrated control system. This field is mainly used to distinguish warehouses or areas. The default value is 1.
	deviceType		int	2	Yes	Fixed value: 0
	deviceCode		string	64	No	Device ID: Can be used for fuzzy search. Device IDs are separated by commas.
Response	code		int	6	Yes	Return status code
	desc		string	64	Yes	Status code description
	data		JsonArray		Yes	Device type information returned
	data	deviceCode	string	32	Yes	Device serial number
		payLoad	string	4	No	Device load status
		orderId	string	100	No	ID of the order in progress from a third party system
		shelfNumber	string	64	No	Code of the shelf the AGV is moving
		devicePostionRec	Array	32	No	X- and Y-coordinates of the device in QR code form. The former value is X-coordinate and the latter value is Y-coordinate.
		devicePosition	string	32	Yes	Device location
		battery	string	4	Yes	Battery power
		deviceName	string	32	Yes	Device name
		deviceStatus	int	2	Yes	Device status:

						0: Offline 1: Idle 2: Fault 3: Initializing 4: In task 5: Charging 7: Upgrading
		state	string	32	Yes	Idle: Idle Initializing: Initializing InTask: In task Fault: Fault Offline: Offline InCharging: Charging InUpgrading: Upgrading
		orientation	Int	6	No	Orientation in 0.001 degrees
		speed	int	6	No	Speed in mm/s

4.1.2.1.3 Example

- Request

```
{
"areaId":"1",
"deviceType":"0"
}
```

Note:

- Successful response

```
{
    "code": 1000,
    "data": [
        {
            "taskPath": "55300593,03200340,55300593",
            "payLoad": "0.0",
            "orderId": "DWCS00160521",
            "shelfNum": "AG02-062",
            "devicePosition": "55300593",
            "devicePostionRec": [
                136771,
                54039
            ],
```

```
                "state": "InTask",
                "deviceCode": "6L0F5E5PA400002",
                "battery": "59",
                "deviceName": "0001",
                "deviceStatus": 4
            }
        ],
        "desc": "success"
}
```


4.1.2.2 Suspend/recover device

4.1.2.2.1 Interface Details

- Interface name: http://IP:7000/ics/out/controlDevice
- Function description: This interface is used by the customer system to stop or restore the AGV.
- Interface protocols: HTTP
- Provider: ICS (integrated control system
- Caller: Third party system

4.1.2.2.2 Parameter Details

Table 4-11 Parameter details of device suspending/restoring interface

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	areaId	Int	10	Yes	Area ID: Provided by the integrated control system. This field is mainly used to distinguish warehouses or areas. The default value is 1.
	deviceNumber	string	64	No	Device ID. The device ID is null if all devices in the area are to be suspended or restored (devices are separated by commas.
	all	Int	2	No	Whether to suspend or restore all devices in the area 0: Not all 1: All
	controlWay	Int	2	Yes	0: Suspend device 1: Restore device
Response	code	int	6	Yes	Return status code

	data	String		Yes	Returned result
	desc	string	64	Yes	Status code description

4.1.2.2.3 Example (suspend part of devices

- Request

```
{
    "areaId": 10,
    "deviceNumber":"device1,device2",
    "all":0,
    "controlWay": 0
}
```

- Response

```
{
    "code": 1000,
"data": ""
"desc": "Request successfully"
}
```

4.1.2.2.4 Example (suspend all devices

```
{
    "areaId": 10,
    "deviceNumber":"",
"all":1,
"controlWay": 0
}
```

- Response

```
{
    "code": 1000,
"data": ""
"desc": "Request successfully"
}
```

4.1.2.3 Cancel device preemption

4.1.2.3.1 Interface Details

- Interface name: http://IP:7000/ics/out/removeAGV

- Function description: This interface is used by a third party system to cancel device preemption.
- Interface protocols: HTTP
- Provider: ICS (integrated control system
- Caller: Third party system
- Application: Applicable to scenarios where a third party system wants to actively release devices. When a third party system does not want to release the device after the specified task ends, while the third party system no longer wants the device to continue the task due to some reason, the third party system will actively release the device to allow it to charge.

4.1.2.3.2 Parameter Details

Table 4-12 Parameter details of device preemption canceling interface

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
	deviceNumber	string	50	Yes	Device ID: Multiple devices are separated by commas.
Response	code	int	6	Yes	Return status code
	data	string		Yes	Return result
	desc	string	64	Yes	Status code description

4.1.2.3.3 Example

- Request

```
{
"deviceNumber":"1112,2223"
    }
```

- Response

```
{
   "code": 1000,
   "data": "",
   "desc": "Request successfully"
}
```

4.1.2.4 Get Algorithm-based Programming Status

4.1.2.4.1 Interface Details

- Interface name: http://IP:PORT/getAGVTaskInfo
- Function description: This interface is called when a third party system wants to get a certain piece of device programming information.
- Interface protocols: HTTP
- Provider: ICS (integrated control system
- Caller: Third party system

- Application: Applicable to in-task scenarios where you want to view AGV information on the PDA.

4.1.2.4.2 Parameter Details

Table 4-13 Parameter details of algorithm-based programming status getting interface

Parameter type	Parameter Name		Field Type	Max. Length	Required or Not	Description
Request	deviceCode		string	64	No	Device serial number
	deviceNumber		string	50	No	Device ID. Pass in either device ID or device SN
Response	code		int	6	Yes	Return status code
	desc		string	64	Yes	Status code description
	data		JsonArray		Yes	Device type information returned
	data	alarmMsg	string	32	No	Alarm message
		data	string	4	No	Material information, reserved field
		targetPositionName	string	32	No	Destination point ID
		advice	string	32	No	Handling advice
		remark	string	4	No	Current path segment status, device status, and algorithm status
		startPositionName	string	32	No	Starting point ID
		taskTypeName	string	2	No	RCS subtask type name
		startPosition	string	32	No	Starting point of current path segment in QR code form
		targetPosition	string		No	Destination point of current path segment in QR code form
		outOrderId	string		No	Order ID from a third party system. The value is valid when a task is being executing
		taskTemplateName	string	128	No	RCS task template name
		state	string		Yes	Device status: 0: Offline 1: Idle 2: Fault 3: Initializing 4: In task 5: Charging 7: Upgrading
		taskId	string		No	RCS task ID

4.1.2.4.3 Example

- Request

```
{
"deviceCode":"7B03A73PA400010"
}
```


Note:

- Response


```
{
    "code": 1000,
    "data": {
        "alarmMsg": "task 24@80937@461103 not completed",
        "data": "[{\"materielNum\":\"\",\"materielName\":\"\",\"materielAccount\":\"\"}]",
        "targetPositionName": "",
        "advice": "check whether the task is completed",
        "remark": "[path segment status: algo24_442_869 executing; algorithm status: in task; normal movement; device status: abnormal status; programming status: programmable status;]",
        "startPositionName": "",
        "taskTypeName": "Shelf/pallet movement (arriving with load",
        "startPosition": "77000005",
        "alarmFlag": 1,
        "targetPosition": "77000005",
        "outOrderId": "",
        "taskTemplateName": "QR code-based warehouse-out sorting",
        "shelfNumber": "99000102",
        "state": 1,
        "taskId": "80938"
    },
    "desc": "success"
}
```


4.1.2.5 Get task path

4.1.2.5.1 Interface Details

- Interface name: http://IP:7000/ics/out/getRobotTaskPath
- Function description: This interface is called by the customer system to get the details of the AGV task path.
- Interface protocols: HTTP
- Provider: ICS (integrated control system

- Caller: Third party system

4.1.2.5.2 Parameter Details

Table 4-14 Parameter details of task path getting interface

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	areaId	int	10	Yes	Area ID. All task path information will be obtained when only this value is passed in.
	deviceNumber	string	64	No	Device ID, the device for which the path must be obtained
	fatherAreaId	int	2	No	Parent area ID in cross-area scenarios
	isCross	boolean	2	No	Cross-area or not, true or false
	deviceCode	string	2	No	Device serial number
Response	code	int	6	Yes	Return status code
	desc	string	64	Yes	Status code description
	areaInfo	jsonArray	64	Yes	Area details
	areaId	string	64	Yes	Area ID
	devices	jsonArray	64	Yes	Device Information
	deviceNum	string	64	Yes	Device ID
	pathNodes	jsonArray	64	Yes	Path details. The four values are QR code value, X-coordinate, Y-coordinate, and angle respectively.
	pathLockRects	jsonArray	64	Yes	Path area range. One set of X-coordinate and Y-coordinate for each two areas
	pathLockCircs	jsonArray	64	Yes	Periphery detection area when the device is running
	deviceCode	string	2	Yes	Device serial number

4.1.2.5.3 Example

- Request

```
{
"areaId": 10,
    "deviceNumber": "1000",
    "deviceCode": "mmC0600000",
    "fatherAreaId": "",
```

```
        "isCross": false}
```


- Response


```
{
  "code": 1000,
        "data": {
              "areaInfo": [
                    { "areaId": 2,
                          "devices": [
                                {
                                      "deviceNum": "1000",
                                      "pathNodes": [
                                            "11000020,2470,10350,999000",
                                            "11000032,3820,10350,999000",
                                            "11000031,3820,9000,999000",
                                            "11000007,1120,9000,999000"
                                      ],
                                      "pathLockRects": [
                                            "4135,9855,3505,9855,3505,10845,4135,10845"
                                      ],
                                      "pathLockCircs": [],
                                      "deviceCode": "mmC0600000"
                                }
                          ]
                    }
              ]
        },
        "desc": "request succeeded"}
```


4.1.3 Bin Related Interface

4.1.3.1 Get Shelf Location

4.1.3.1.1 Interface Details

- Interface name: http://IP:7000/ics/out/shelf/getShelfPosition

- Function description: This interface is called by a third party system to get the shelf location maintained in the RCS.
- Interface protocols: HTTP
- Provider: ICS (integrated control system
- Caller: Third party system
- Application: When the shelf location is maintained in the RCS and all shelf location information should be synchronized to the MES and WMS

4.1.3.1.2 Parameter Details

Table 4-15 Parameter details of shelf location getting interface

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	areaId	String	10	Yes	Area ID. All shelves in this area are queried when only this value is passed in.
	shelfNums	string	100	No	Collection of shelf codes which are separated with commas.
Response	shelfNum	string	10	Yes	Shelf code
	shelfCurrStation	string	10	Yes	QR code-based shelf location. If the shelf has not registered location information to the RCS system, -1 is returned and the following field is null.
	posX	int	10	No	X-coordinate of shelf
	posY	Int	10	No	Y-coordinate of shelf
	shelfType	Int	6	No	Shelf type
	angle	string	10	No	Shelf angle

4.1.3.1.3 Example

- Request

```
{
    "areaId":"3",
    "shelfNums":"shelf01,test"
}
```

- Response

```
{
    "code": 1000,
```

```
        "data": [
            {
                "angle": "-1000",
                "posX": -1000000,
                "posY": -1000000,
                "shelfCurrStation": "-1",
                "shelfNum": "shelf01",
                "shelfType": 1
            },
            {
                "angle": "0",
                "posX": 12160,
                "posY": 14400,
                "shelfCurrStation": "01000015",
                "shelfNum": "test",
                "shelfType": 1
            }
        ],
        "desc": "Request successfully"
}
```


4.1.3.2 Change Shelf Location

4.1.3.2.1 Interface Details

- Interface name: http://IP:7000/ics/out/shelf/changePosition
- Function description: This interface is called to change the shelf location.
- Interface protocols: HTTP
- Provider: ICS (integrated control system
- Caller: Third party system
- Application: When the shelf location is maintained in the RCS and the location of a shelf is manually changed. Information is synchronized to the RCS

4.1.3.2.2 Parameter Details

Table 4-16 Parameter details of shelf location updating interface

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	areaId	int	10	Yes	Area ID: Provided by the integrated control system. This field is mainly used to distinguish warehouses or areas.
	shelfNum	string	10	Yes	Shelf code
	destPosition	string	10	Yes	Destination point to bind to
	action	Int	2	Yes	1: Bind shelf 2: Unbind shelf
Response	code	int	6	Yes	Return status code
	data	JsonObject		Yes	Return result
	desc	string	64	Yes	Status code description

4.1.3.2.3 Example

- Request

```
[
    {
        "areaId": 1,
        "destPosition": "66000079",
        "shelfNum": "s001",
        "action": 1
    }
]
```

- Response

```
{
    "code": 1000,
    "data": {},
    "desc": "Request successfully"
}
```

4.1.3.3 Get all bin information

4.1.3.3.1 Interface Details

- Interface name: http://IP:7000/ics/out/getStockStatus

- Function description: In some cases, the RCS does not record shelf location information. The RCS may determine whether there is a shelf in a certain location according to task execution status. The RCS defines this as bin information. These information can be obtained by a third party system through this interface.
- Interface protocols: HTTP
- Provider: ICS (integrated control system
- Caller: Third party system
- Application: When the RCS records full or empty bin status and the bin status will be displayed in the WMS and MES

4.1.3.3.2 Parameter Details

Table 4-17 Parameter details of bin information getting interface

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	areaId	string	10	Yes	Area ID. This is used to distinguish the workshop from which the bin information is obtained when there are multiple areas. Enter 1 when there is only one area.
	pageSize	string	6	No	Query 20 entries each time when it is left blank
	pageNo	string	6	No	Query page 1 each time when it is left blank
	areaName	string	12	No	Name of line logistics. This can be queried on the third party area configuration page in the RCS.
Response	code	int	6	Yes	Return status code
	desc	string	64	Yes	Status code description
	data	JsonObject		Yes	Response
	pageSize	Int	5	No	Size per page
	pageNo	Int	5	No	Page
	totalPage	Int	5	No	Total pages
	totalNum	Int	10	No	Total counts
	stockList	JSONArray			Collection of bin information
	areaId	Int	10	Yes	Area ID of this bin
	inTask	Int	2	Yes	Whether a task is triggered at this point
	qrContent	String	32	Yes	Bin QR code
	stockStatus	Int	2	Yes	Bin status. In general, 0 means empty and 2 means full

4.1.3.3.3 Example

- Request

```
{
```

```
    "areaId":"1",
    "pageSize":"2",
    "pageNo":"1",
    "areaName":"FC05"
}
```


- Response


```
{
    "code": 1000,
    "data": {
        "stockList": [
            {
                "areaId": "1",
                "inTask": "0",
                "qrContent": "80000005",
                "stockStatus": 0
            }
        ],
        "totalNum": 2,
        "totalPage": 1,
        "pageNo": 1,
        "pageSize": 2
    },
    "desc": "success"
}
```


4.1.3.4 Change Bin Status

4.1.3.4.1 Interface Details

- Interface name: http://IP:7000/ics/stock/update/appStockStatus
- Function description: This interface is called to manually synchronize bin change information to the RCS upon manual intervention of shelf location from a third party system.
- Interface protocols: HTTP
- Provider: ICS (integrated control system
- Caller: Third party system

- Application: The RCS has maintained empty and full status at some points according to task status even when shelves are not maintained. After the shelf location is manually changed, you have to manually maintain and change this data and synchronize the change to the RCS.

4.1.3.4.2 Parameter Details

Table 4-18 Parameter details of bin status updating interface

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	qrContent	string	32	Yes	RCS-assigned QR code for bin
	nodeStatus	string	2	Yes	Bin status: 0 - empty bin 2 - full bin
Response	code	int	6	Yes	Return status code
	desc	string	64	Yes	Status code description

4.1.3.4.3 Example

- Request

- {
- "qrContent":"80003212",
- "nodeStatus":"0"
- }
- Note:
  - Successful response

- {
- "code":1000,
- "desc":"success"
- }

4.1.4 Traffic Control Related Interface

4.1.4.1 Apply for entering the traffic control area

4.1.4.1.1 Interface Details

- Interface name: http://IP:7000/ics/out/askZone
- Function description: This interface is called when other manufacturer's AMR needs to pass the area where the current AMR is operating, the other manufacturer's AMR will make an application, and the RCS will start traffic control.
- Interface protocols: HTTP
- Provider: ICS (integrated control system

- Caller: Third party system

4.1.4.1.2 Parameter Details

Table 4-19 Parameter details of the interface to apply for entering the traffic control area

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	reqCode	string	10	Yes	Request ID. One-pass request uses the same ID. This ID can be used to make several pass requests when several AMRs apply for passing through the traffic control area.
	deviceNumber	string	64	Yes	AMR ID
	askZoneCode	string	10	Yes	ID of traffic control area to pass through
	action	int	1	Yes	1: Request for entrance 2: Request for departure
Response	code	int	6	Yes	Return status code. No entry when no 1000 is returned
	desc	string	64	Yes	Status code description
	data	JSON	64	No	

4.1.4.1.3 Example

- Request

```
{
   "reqCode": "1235435653",
   "deviceNumber":"1001"，
   "askZoneCode":"MAREA12",
   "action": 1
}
```

- Response

```
{
   "code": 1000,
   "desc": "Request successfully"
}
```

4.1.4.2 Enable and disable the prohibited region

4.1.4.2.1 Interface Details

- Interface name: http://IP:7000/ics/out/enableForbiddenZone

- Function description: This interface is called when the customer system needs to enable or disable the prohibited region.
- Interface protocols: HTTP
- Provider: ICS (integrated control system
- Caller: Third party system

4.1.4.2.2 Parameter Details

Table 4-20 Parameter details of prohibited region enabling/disabling interface

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	indBind	int	1	Yes	Enable/disable prohibited region: 1: Enable prohibited region, 0: Disable prohibited region
	matterArea	string	64	Yes	Name of prohibited region
Response	code	int	6	Yes	Return status code. Modification failed when no 1000 is returned
	desc	string	64	Yes	Status code description
	data	JSON	64	No	

4.1.4.2.3 Example

- Request

```
{
    "indBind": 1,
    "matterArea":"prohibited region 1"
}
```

- Response

```
{
    "code": 1000,
    "desc": "Request successfully"
}
```

4.1.5 Map Related Interface

4.1.5.1 Get the map

4.1.5.1.1 Interface Details

- Interface name: http://IP:7000/ics/out/topologyList
- Function description: This interface is called when the customer system needs to get map files. A map download URL is returned.
- Interface protocols: HTTP

- Provider: ICS (integrated control system
- Caller: Third party system

4.1.5.1.2 Parameter Details

Table 4-21 Parameter details of prohibited region enabling/disabling interface

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	areaId	int	10	Yes	Area ID
Response	code	int	6	Yes	Return status code. Modification failed when no 1000 is returned
	desc	string	64	Yes	Status code description
	data	JSON	64	No	JSON object
	mapZipUrl	string	5000	No	Download URL of compressed map file
	mapJsonUrl	string	5000	No	Download URL of map file in JSON format
	mapId	int	10	No	Map ID, same as the area ID

4.1.5.1.3 Example

- Request


```
{
    "areaId": 8
}
```


- Response


```
{
    "code": 1000,
    "data": {
        "mapZipUrl": "http://10.55.216.75:8888/textureMap/8/default/Map/commonTopo.zip?token=3ndR1SMbDF70DRzS3b0jSD8KafHVxrzE",
        "mapId": 8,
        "mapJsonUrl": "http://10.55.216.75:8888/textureMap/8/default/Map/compress.json?token=ZtEs6cWb54jVnabmfZoC623bBDR9Vw9J"
    },
    "desc": "Request successfully"
}
```

4.2 Interfaces Provided by Third Party System

- The ICS system provides the function of active reporting and a third party system provides the interface URL to the site engineer. This URL will be configured in the ICS system, and the third party system will implement this interface based on the interface protocol to implement mutual data interaction.

4.2.1 Reporting Device Status to Third Party System

4.2.1.1 Interface Details

- Interface name: Provided by a third party system
- Function description: This interface is provided by a third party system. This interface is called from a third party system when the integrated control system needs to actively report the AGV status to the third party system.
- Interface protocols: HTTP
- Provider: Third party system
- Caller: ICS (integrated control system
- Application: When a third party needs to set a data dashboard
- Calling logic: Report in polling mode without retry

4.2.1.2 Parameter Details

Table 4-22 Parameter details of AGV status

Request parameters	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	deviceCode	string	32	Yes	Device serial number
	payLoad	string	4	No	Device load status
	devicePostionRec	Array	32	No	X- and Y-coordinates of the device in QR code form. The former value is X-coordinate and the latter value is Y-coordinate.
	devicePosition	string	32	Yes	Device location
	battery	string	4	Yes	Battery power
	deviceName	string	32	Yes	Device name
	deviceStatus	int	2	Yes	Device status: 0: Offline 1: Idle 2: Fault 3: Initializing 4: In task 5: Charging 7: Upgrading
	state	String	32	Yes	Idle: Idle

					Initializing: Initializing InTask: In task Fault: Fault Offline: Offline InCharging: Charging InUpgrading: Upgrading
	orientation	int	6	No	Orientation in 0.001 degrees
	orderId	string	100	No	ID of the order in progress from a third party system
	shelfNumber	string	64	No	Code of the shelf the AGV is moving
	speed	int	6	No	Speed in mm/s
Response	code	int	6	Yes	Return status code
	desc	string	64	Yes	Status code description

4.2.1.3 Example

- Request

```
[
    {
      "taskPath": "01000037,01000040",
      "payLoad": "1.0",
      "orderId": "5975858_test_15_test36",
      "shelfNum": "TEST111",
      "devicePosition": "01001045",
      "devicePostionRec": [5410, 2250],
      "shelfNumber": "TEST111",
      "state": "InTask",
      "deviceCode": "ffff0000",
      "battery ": "27 ",
      "deviceName ": "ffff ",
      "deviceStatus ": 4
}
 ]
```

- Response

```
{
   "code": 1000,
   "desc": "Request successfully"
}
```

4.2.2 Requesting Third Party System to Get Task Point In a Task

4.2.2.1 Interface Details

- Interface name: Provided by a third party system
- Function description: This interface is to be provided by a third party system. The integrated control system makes an request to a third party system to get a task point and the third party system returns one task point.
- Interface protocols: HTTP
- Provider: Third party system
- Caller: ICS (integrated control system
- Application: Applicable to scenarios where the task of the AGV is to move goods from point A to point B, but only point A is assigned when the task is assigned, and the AGV makes a request to get point B after arriving at point A.

4.2.2.2 Parameter Details

Table 4-23 Parameter details of task point getting interface

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	orderId	string	100	Yes	Order ID from a third party system
	modelProcessCode	string	100	Yes	Business process template number
Response	code	int	6	Yes	Return status code
	desc	string	64	Yes	Status code description
	data	JsonObject		Yes	Task point information returned
	pointName	string	10	Yes	Task point returned. If an area is returned, select matching points in the RCS system.

4.2.2.3 Example

- Request

```
{
"orderId":"task1111",
"modelProcessCode":"kongkuangfanhui"
}
```

- Response

```
{
"code":1000,
```

```
"desc":"success",
"data":{
        "pointName":"pointA"
}
}
```

4.2.3 Report alarm message

4.2.3.1 Interface Details

- Interface name: Provided by a third party system
- Function description: This interface is provided by a third party system. The interface is called when the integrated control system needs to actively report the alarm message.
- Interface protocols: HTTP
- Provider: Third party system
- Caller: ICS (integrated control system
- Application: When a third party system needs to display the AGV alarm messages in the RCS to the MES and WCS systems.
- Calling logic: Report once when an alarm is triggered. No retry

4.2.3.2 Parameter Details

Table 4-24 Parameter details of alarm message

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Request	deviceNum	string	6	No	Device ID
	alarmDesc	string	500	Yes	Alarm description
	alarmType	int	6	Yes	Alarm type, see Appendix 3
	areaId	int	2	Yes	Area ID
	alarmReadFlag	int	1	Yes	Read or not
	channelDeviceId	string	32	No	Alarm source
	alarmSource	string	6	Yes	Alarm source system
	channelName	string	256	No	Suggestion
	alarmDate	string	15	Yes	Alarm date
	deviceName	string	15	No	Device SM, generated from the RCS
	alarmGrade	int	2	Yes	Alarm priority: 0, 1, 2, The higher the number, the more critical the alarm
Response	code	int	6	Yes	Return status code
	desc	string	64	Yes	Status code description

4.2.3.3 Example

- Request


```
{
    "deviceNum": "jbz1",
    "alarmDesc": "device offline",
    "alarmType": 1,
    "areaId": 13,
    "alarmReadFlag": 1,
    "channelDeviceId": "80000292",
    "alarmSource": "rcs",
    "channelName": "1. Check network connection \\r\\n 2. Restart the device",
    "alarmDate": 538227704,
    "deviceName": "jbztest0000",
    "alarmGrade": 2
}
```


- Response


```
{
   "code": 1000,
   "desc": "Request successfully"
}
```


4.2.4 Actively report shelf location to a third party system

4.2.4.1 Interface Details

- Interface name: Provided by a third party system
- Function description: This interface is called to actively get location information.
- Interface protocols: HTTP
- Provider: Third party system
- Caller: ICS (integrated control system
- Application: When shelf locations are fully maintained in the RCS and the MES and WMS want real-time shelf location information from the RCS
- Calling logic: Report in polling mode without retry

4.2.4.2 Parameter Details

Table 4-25 Parameter details of shelf location actively getting interface

Parameter type	Parameter Name	Field Type	Max. Length	Required or Not	Description
Full-parameter request	shelfNum	string	10	Yes	Shelf code
	shelfCurrStation	string	10	Yes	QR code-based shelf location. If the shelf has not registered location information to the RCS system, -1 is returned and the following field is null.
	posX	int	10	No	X-coordinate of shelf
	posY	Int	10	No	Y-coordinate of shelf
	shelfType	Int	6	No	Shelf type
	angle	string	10	No	Shelf angle
Response	code	int	6	Yes	Return status code
	data	JsonObject		Yes	Return result
	desc	string	64	Yes	Status code description

4.2.4.3 Example

- Request

```
[{
    "shelfNum":"66000454,
    "shelfCurrStation": "66000079",
    "posX": 32434
    "posY":12313
    "shelfType":1,
    "angle":0
},
]
```

- Response

```
{
   "code": 1000,
   "desc": "Request successfully"
}
```

Section 5 Interactive Flow Chart

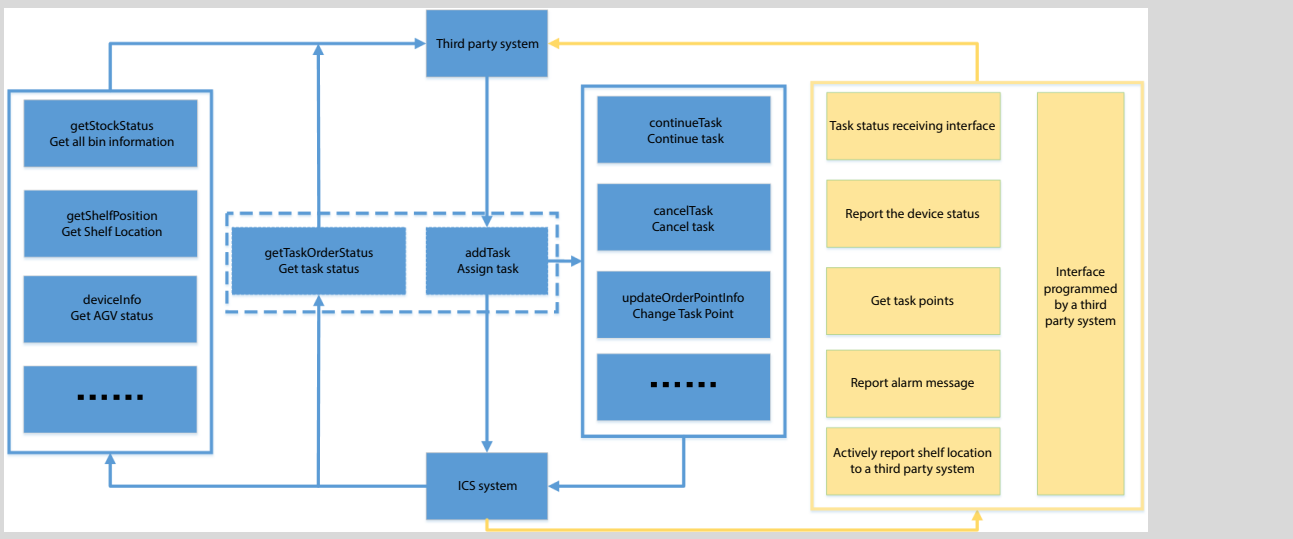
5.1 Basic Principle

- For a QR code-based navigation system, the map is actually a collection of QR code values. QR code values represent coordinates in the map and can also represent locations where shelves are placed. QR code values can be named to represent actual applied locations. The name can contain letters and numbers, for example:
- QR code value ————————→ Named value

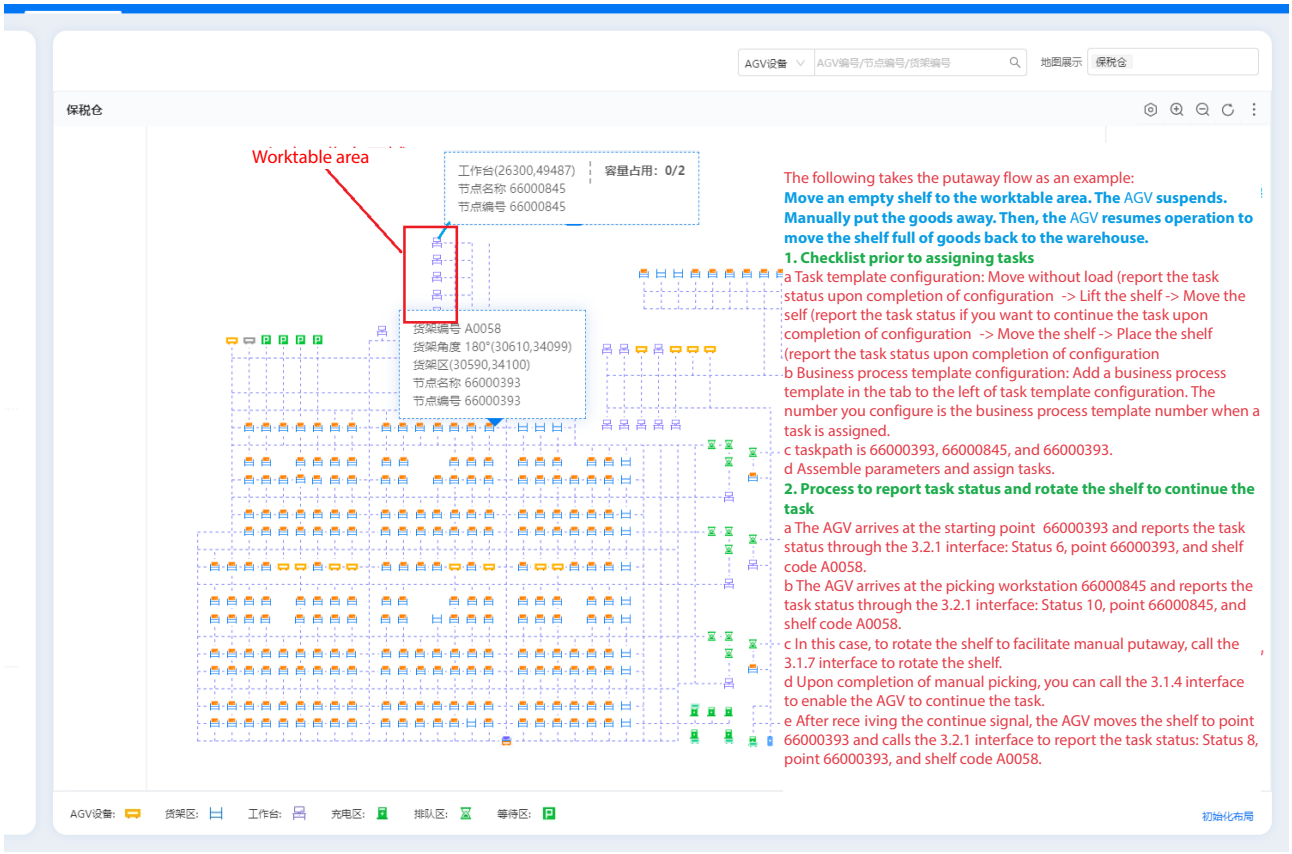
  66000393 BEN00001

- For a laser-based navigation system, the map is actually a collection of coordinate values. The coordinates in the map represent locations and can also represent locations where shelves are placed. Coordinate values can be named to represent actual applied locations. The name can contain letters and numbers, with a length of 4 to 10 digits.
- Task template: An AGV is a driven device. The AGV acts based on task template configurations section by section when executing a task. A task template is used to configure these subtask actions.
- Business template: A business template is prepared based on business scenarios. That is, a business template is a collection of task templates and contains subtasks throughout the entire business. Therefore, modelProcessCode corresponds to business template.
- The AGV moves from a coordinate point to another one (QR code or laser coordinate throughout the entire business, and then executes corresponding actions at the coordinate point (lifting equipment: lifting operation, rolling equipment: transfer operation. A business template actually abstracts AGV movement services into specific tasks, and after these tasks are assigned, the AGV executes corresponding actions.

5.2 Interface Calling Relationship Diagram



5.3 Schematic Diagram of Interaction Flow



The following takes the putaway flow as an example:

Move an empty shelf to the worktable area. The AGV suspends. Manually put the goods away. Then, the AGV resumes operation to move the shelf full of goods back to the warehouse.

1. Checklist prior to assigning tasks

a Task template configuration: Move without load (report the task status upon completion of configuration -> Lift the shelf -> Move the self (report the task status if you want to continue the task upon completion of configuration -> Move the shelf -> Place the shelf (report the task status upon completion of configuration

b Business process template configuration: Add a business process template in the tab to the left of task template configuration. The number you configure is the business process template number when a task is assigned.

c taskpath is 66000393, 66000845, and 66000393.

d Assemble parameters and assign tasks.

2. Process to report task status and rotate the shelf to continue the task

a The AGV arrives at the starting point 66000393 and reports the task status through the 3.2.1 interface: Status 6, point 66000393, and shelf code A0058.

b The AGV arrives at the picking workstation 66000845 and reports the task status through the 3.2.1 interface: Status 10, point 66000845, and shelf code A0058.

c In this case, to rotate the shelf to facilitate manual putaway, call the 3.1.7 interface to rotate the shelf.

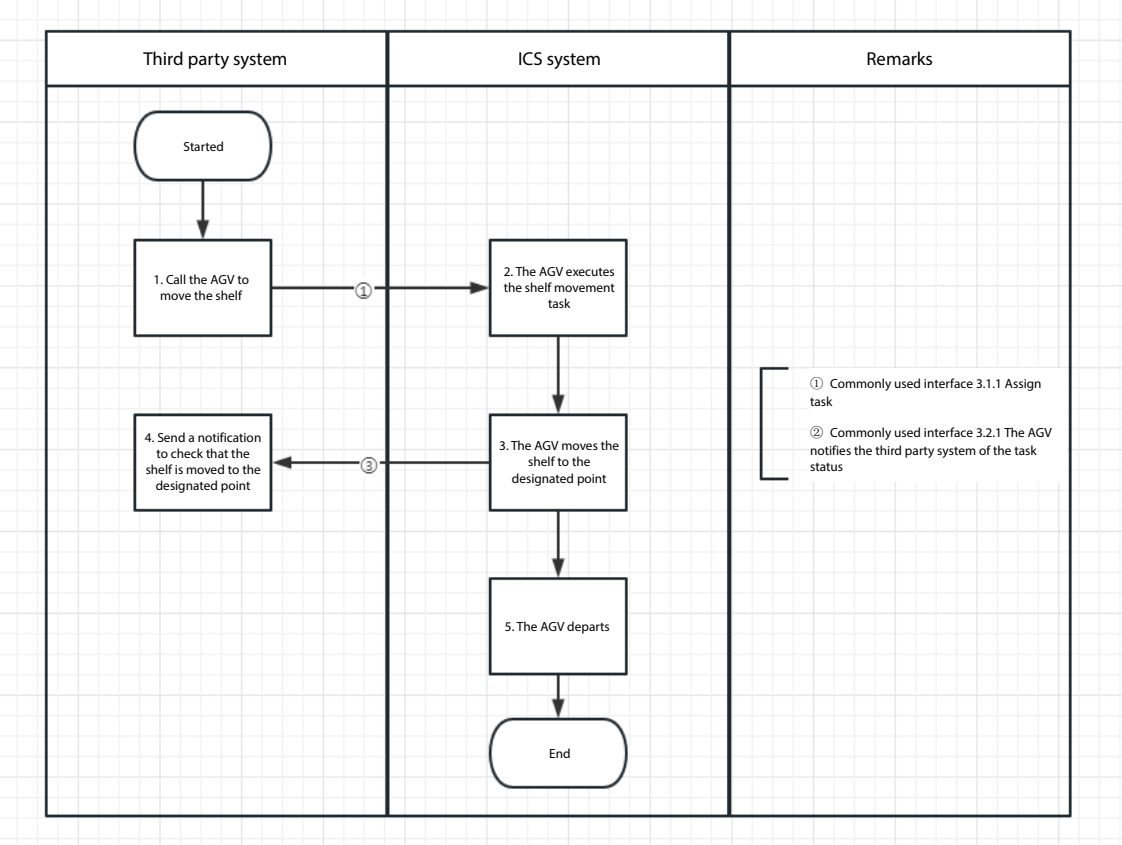
d Upon completion of manual picking, you can call the 3.1.4 interface to enable the AGV to continue the task.

e After rece iving the continue signal, the AGV moves the shelf to point 66000393 and calls the 3.2.1 interface to report the task status: Status 8, point 66000393, and shelf code A0058.

5.4 Typical Scheduling Scenario

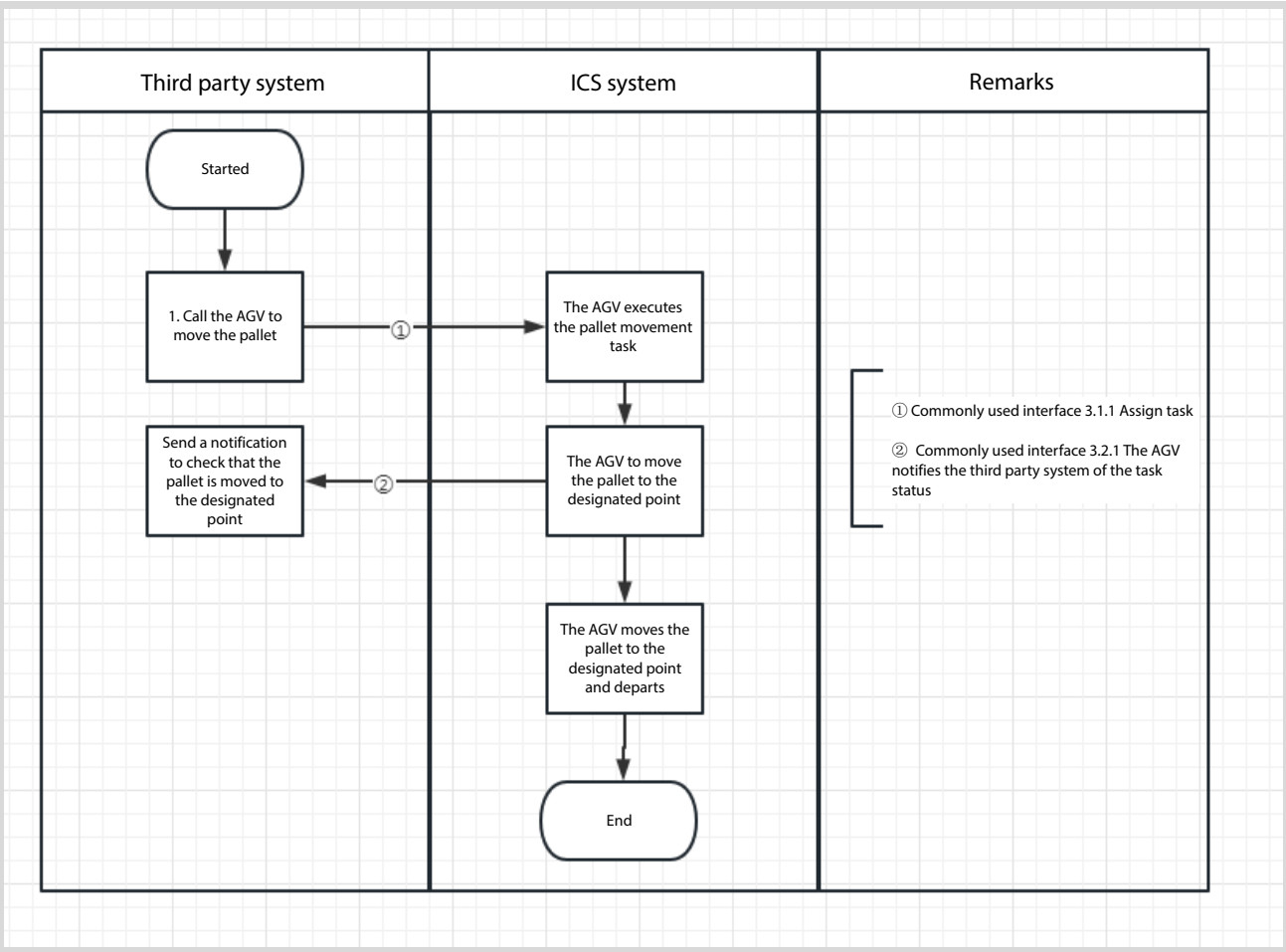
5.4.1 Scenario 1: General shelf movement scenario

Table 5-1 General shelf movement scenario



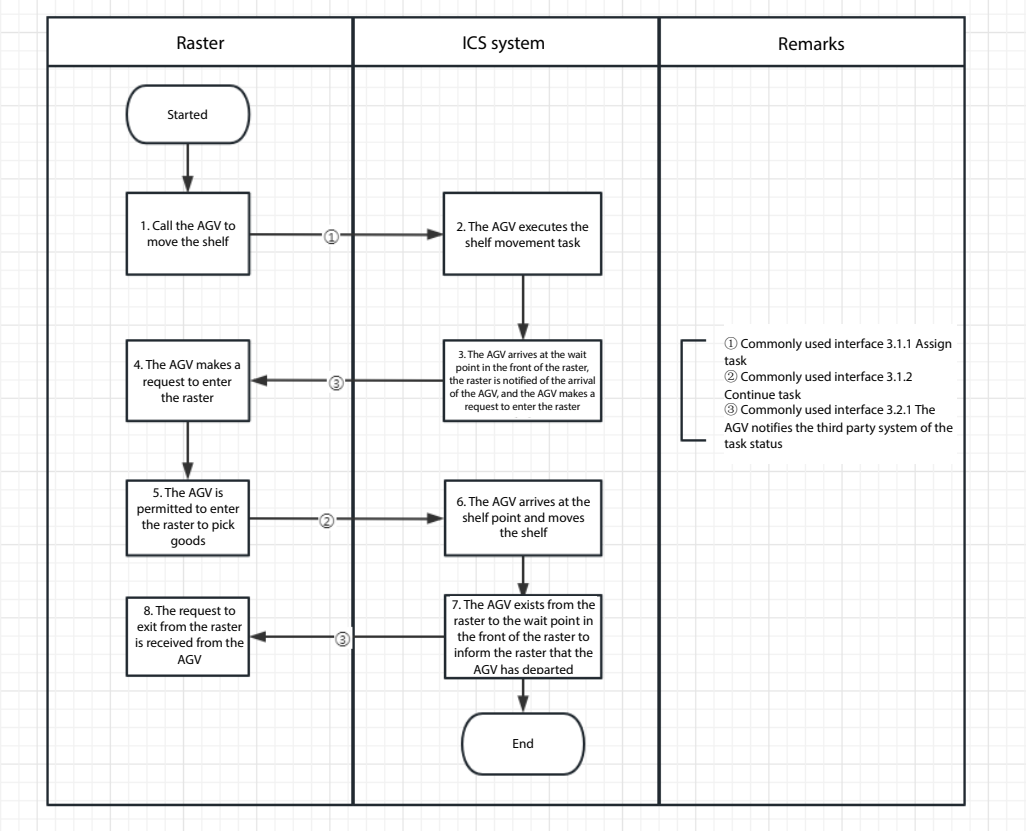
5.4.2 Scenario 2: General pallet movement with forklift scenario

Table 5-2 General pallet movement with forklift scenario



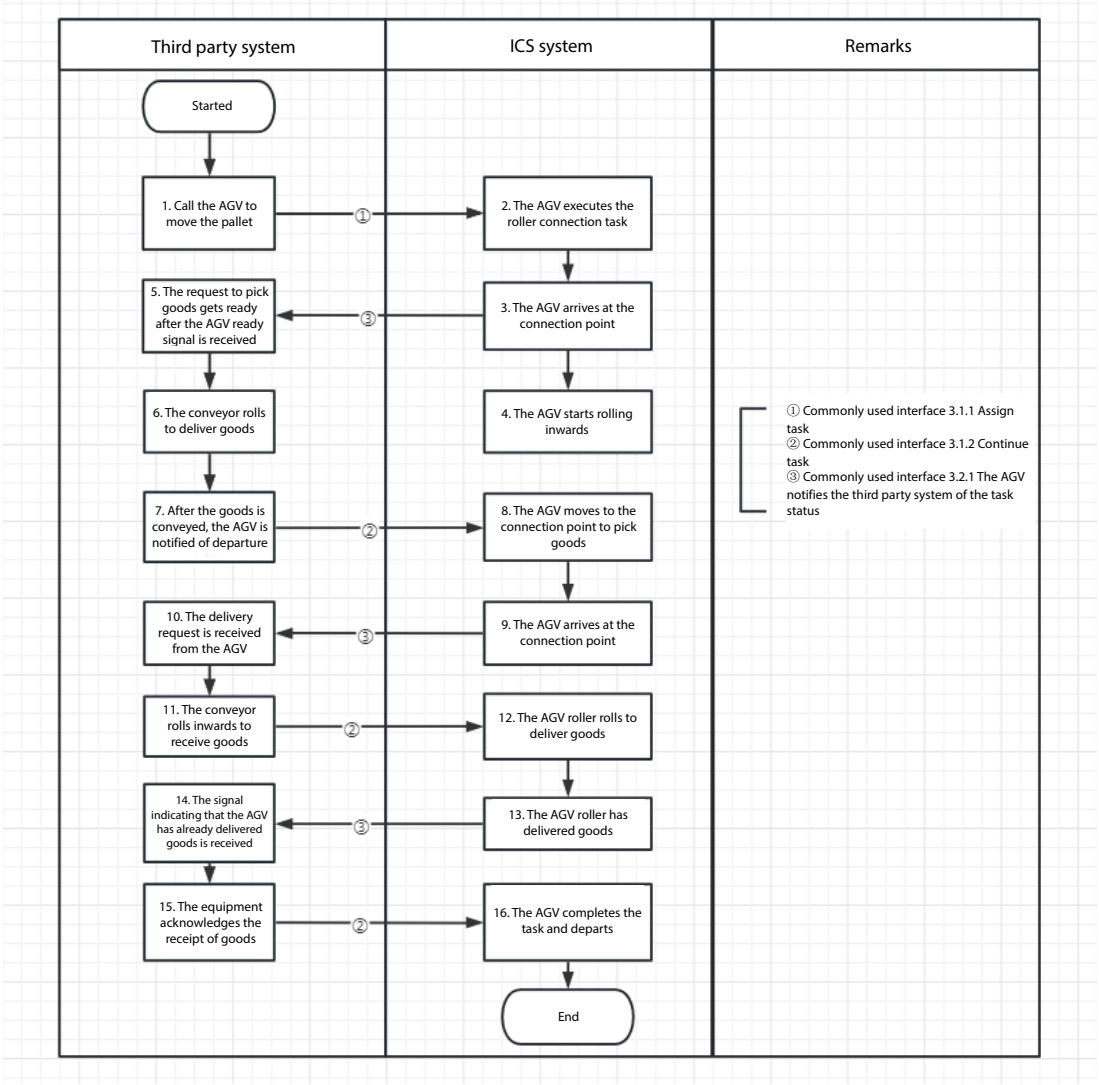
5.4.3 Scenario 3: Movement in and out of raster

Table 5-3 Movement in and out of raster



5.4.4 Scenario 4: Movement with roller scenario

Table 5-4 Movement with roller scenario



Suggestions on Third Party Development

- General application requirements can be easily met as long as a task is assigned and task status is reported. Other interfaces can be called as needed.

Appendix 1 Legal Information

Declaration of Trademark

- VGA is the trademark of IBM.
- Windows mark and Windows are trademarks or registered trademarks of Microsoft.
- Other trademarks or company names that may be referred to in this document are properties of their respective owners.

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Appendix 2 Cybersecurity Recommendations

- Essential measures for ensuring basic network security for the device:

1. Use A Strong Password

Refer to the following recommendations for setting up the password:

- The length should not be less than 8 characters.
- A strong password is comprised of at least two of these three character types: letters in upper and lower cases, numbers, and special characters.
- Do not contain the username in its forward or reverse sequence.
- Do not use continuous characters, such as 123 and abc.
- Do not use repeated characters, such as 111 and aaa.

Timely Update Firmware and Third Party Software

- As is standard procedure in the tech-industry, the firmware of devices should be timely updated to the latest version to ensure the system is current with the latest security patches and fixes. With the device connected to the Internet, it is recommended to turn on automatic online update detection to timely get firmware updates launched by providers.
- It is recommended to download and use the latest third party software.

- Recommended measures to enhance cybersecurity:

Physical Protection

Physical protection is recommended for devices (especially storage devices, such as placing them in a special room, or special cabinet, and well conduct access control and key control to prevent damages to hardware and external devices (USB flash drive, serial ports by unauthorized persons through physical contact.

Changing Passwords Regularly

Regularly change the passwords to your devices to help ensure that only authorized users are able to access the system.

Timely Set up and Update Password Reset

This device supports password reset. To reduce the risk of this function being attacked by hackers, it is recommended to timely setup password reset, including reserved phone number/email, and security questions. If any of such information changes, update the password reset in time. When setting up security questions, set up an answer that is difficult for someone to guess their way into your system.

Enable Account Lock-Down

Account lock-down is enabled by default. To keep your Account safe, it is recommended to keep this setting. After hackers fail multiple attempts to hack passwords, the corresponding Account and source IPs are locked.

Change HTTP and Other Default Service Ports

These ports can be changed to any set of numbers between 1024–65535. Avoiding the default ports reduces the risk of hackers being able to guess which ports you are using.

Enable HTTPS

Enable HTTPS to access Web services through safe channels.

Bind MAC Address

Bind the IP and MAC address of the gateway device to reduce the risk of ARP deception.

Reasonably Allocate Account and Permission

Add users properly according to business and management needs, and allocate minimum sets of permissions to them.

Turn off Unnecessary Services and Use Safe Modes

If not needed, SNMP, SMTP and UPnP can be turned off to reduce the risks for the device.

If necessary, it is strongly recommended to use safe modes, including but not limited to:

- SNMP: Select SNMP v3 and set a stronger encryption password and authentication password.
- SMTP: Select TLS to access the mailbox server.
- FTP: Select SFTP and set a complex password.
- AP hotspot: Select WPA2-PSK encryption mode and set a complex password.

Audio/Video Encryption Transmission

If your audio/video data includes important or sensitive contents, use the encrypted transmission to make it more difficult for someone to steal your data.

Security Audit

- Review online users: Do this from time to time to find out illegal login, if any.
- Review device log: This gives you the IP data of a device trying to log in and the critical operation information of users already in the login status.

Network Log

The limited storage capacity of the device cannot store many logs for a long time. If you need to save logs for a long, enable online log to ensure key logs are synced to the online log server for easy backtracking.

Building Safe Network Environment

To keep the device safer and reduce network risks, we recommend you to:

- Turn off the port mapping of the router to prevent external networks from directly accessing the services of router intranet devices.
- Partition and isolate the network according to the actual network needs. If there is no communication demand between the two subnets, it is recommended to use a VLAN, gatekeeper, etc., to partition the network for network isolation.
- Build up the 802.1x access authentication to reduce the risk of illegal terminals connecting with your private networks.
- Enable device IP/MAC address filtering to limit the hosts that can access the device.

Appendix 3 Alarm Type Reference Table

Alarm Type	Description
1	Device offline
2	Device failure
3	Low battery alarm
4	AMR failure
5	Upgrade failure

6	Node failure
8	Navigation failure
11	Map update failure
12	Obstacle alert
15	High temperature
20	Charging pole failure
21	Charging failure
22	Shelf exception
23	Abnormal path
24	Configuration error
25	Abnormal load
26	Abnormal bin
28	AGV collision
29	Laser sensor failure
30	Goods too high alarm
31	Error when replacing equipment
40	Abnormal RFID
41	Abnormal access control system
42	Abnormal elevator controller
43	Abnormal bin sensor
44	Abnormal smart camera
45	Wait for acknowledgment
46	Service exception
48	Abnormal pager
49	Abnormal industrial-grade indicator
50	Abnormal identification

Appendix 4 Subtask Type Reference Table

Subtask Type	Description
1	No-load movement
2	Shelf/pallet movement (arriving with load
3	Rolling (loading
4	Lifting shelf
5	Place the shelf
10	Goods movement
12	Uncover/reset
13	No-load movement (arriving at the preparation point, without height adjustment
14	Rolling (unloading
16	Picking workbin
17	Placing workbin
18	Hook
19	Unhook

20	No-load movement (arriving at the preparation point, including height adjustment
21	Pallet coordination (pallet stacker AMR
22	Pallet movement (including lifting while walking and lowering while walking
23	Pallet placement (pallet mover AMR
24	Pallet placement (pallet stacker AMR
26	Fork lowering
28	Pallet coordination (pallet mover AMR
29	Shelf recognition
31	Pallet movement (lifting while walking
32	Pallet movement (lowering while walking
33	Shelf/pallet/workbin movement (device arrives
37	Fork height adjustment
38	Rolling (without sensor
43	Workbin/pallet placement
45	Rolling (picking and loading
46	Rolling (picking and unloading
47	Manual intervention (picking and loading
48	Manual intervention (picking and unloading
49	Pallet lifting (pallet mover AMR
57	Pallet placement (pallet stacker AMR
58	Workbin placement (Coaxial forklift AMR
59	No-load movement (picking workbin while walking
62	Hooking
63	Unhooking
66	Workbin/pallet placement V2
68	Multi-model pallet coordination (pallet stacker AMR
69	Multi-model pallet coordination (pallet mover AMR
70	Multi-model shelf recognition